25 August 2026

To the Editors
Applied Mathematics and Computation

Dear Editors,

We submit **“Delegation cascades in evolutionary games: stable computation with strategic depth”** for consideration as a research article in *Applied Mathematics and Computation*. The manuscript studies a finite-population evolutionary game in which delegation depth is a strategic coordinate. It fits the journal’s computational scope by combining an analysed numerical method, Markov-chain dynamics and a systems-oriented application.

The direct mutation-selection chain has about 3.0 x 10^27 population states at the reported parameters. We avoid constructing it by summing the interaction tensor over the horizon law, reducing the rare-mutation process to an embedded chain on 28 designs, and evaluating each fixation probability in O(Z) operations with maximum-shifted exponentials. A closed-class condition handles underflow in extreme transitions. Five complementary checks compare the interaction tensor with Monte Carlo, shifted arithmetic and complete stationary output with 60-digit references, the fixation formula with EGTtools, and the rare-mutation reduction with a tractable full chain.

The analysis proves that realised harm saturates with depth while attributed harm carries a vanishing per-layer factor. Hence every positive attribution retention below one eventually creates a liability shelter: another hand-off lowers the principal’s charge without lowering realised harm. An eventually binding attribution floor closes this shelter. All nine numbered results state their hypotheses and include proofs.

Numerically, transmission erosion and attenuated attribution separately yield unsafe frequencies of 0.018 and 0.011 but jointly yield 0.322; interaction accounts for 91% of the joint effect. Over its attainable range on the reported grid, a statutory attribution floor closely traces the depth-cap safety-social-payoff frontier, while transmission improvement is non-monotone because selection changes the issued intent.

The present manuscript has not been peer reviewed or formally published and is not under consideration elsewhere. A non-peer-reviewed development manuscript source has been publicly accessible in the cited GitHub repository since 20 August 2026. It is a preprint, not a version of record. Two related but non-duplicate companion manuscripts by overlapping author teams are separate submissions in the same coordinated submission round: *Safe designs, unsafe ecosystems: deployment-layer selection in an evolutionary AI race* and *Forgeable greenbeards: multistability and hollow collapse in certified agent populations*. They share the reduced four-strategy interaction layer used as a common benchmark, but study different structural coordinates, mechanisms, state spaces, analytical results and policy questions. Any repeated benchmark quantities are common inputs rather than shared findings. We disclose the companion manuscripts so that the editor can assess the boundaries between the three studies, and we will provide copies on request.

All five authors approved the manuscript and have no competing interests. Code, generated tables, numerical benchmarks and the figure pipeline are available under the MIT licence at https://github.com/trungkiet2005/delegation-cascade.

Thank you for considering the manuscript.

Yours sincerely,

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